AB162x Series EVK Users Guide

Version: 1.0

Release date: 30 Jun 2025

Use of this document and any information contained therein is subject to the terms and conditions set forth in Exhibit 1. This document is subject to change without notice.

Document Revision History

Revision	Date	Description
1.0	30 Jun 2025	Initial version.

Table of Contents

List of Figures

List of Tables

1 Introduction

AB162x is an advanced single-chip solution which integrates baseband and radio for intensive BLE and Zigbee applications. Users can evaluate and configure each function of the AB162x, including GPIO, UART, SPI, I2C / I3C and USB with the Evaluation Kit (EVK).
The EVK serves as a typical Bluetooth data device designed for function evaluation and debugging. Each function can be configured, and parameters can be adjusted using the Airoha Configuration Tool.
The EVK connects with the Airoha Test Control Board (TCB) to perform mass production calibration and functional tests with an MP Tool.

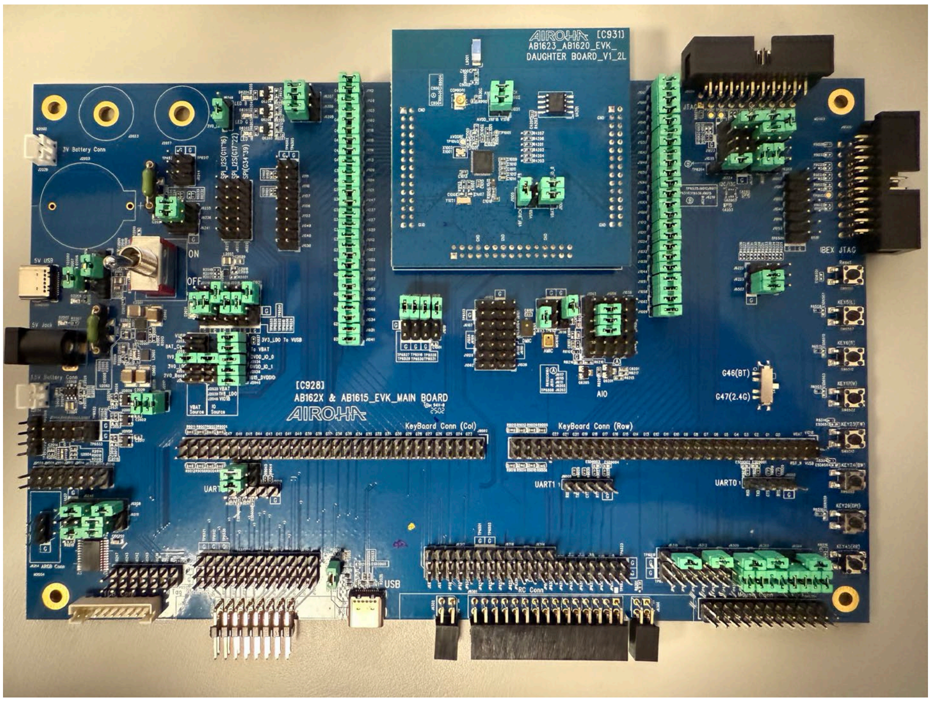

Figure 1-1. Top View of EVK Main Board

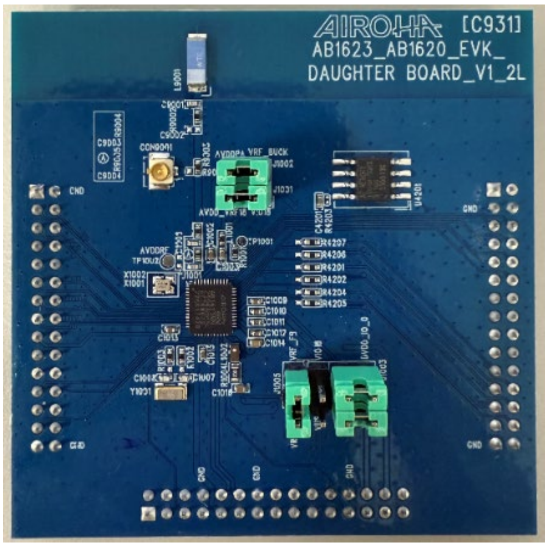

Figure 1-2. Top View of EVK Daughter Board

2 Interface

The EVK provides designers with several I/O and connector interfaces for evaluation. Figure 2-1 shows the top view of the EVK board. Table 1 shows the major components and interfaces.

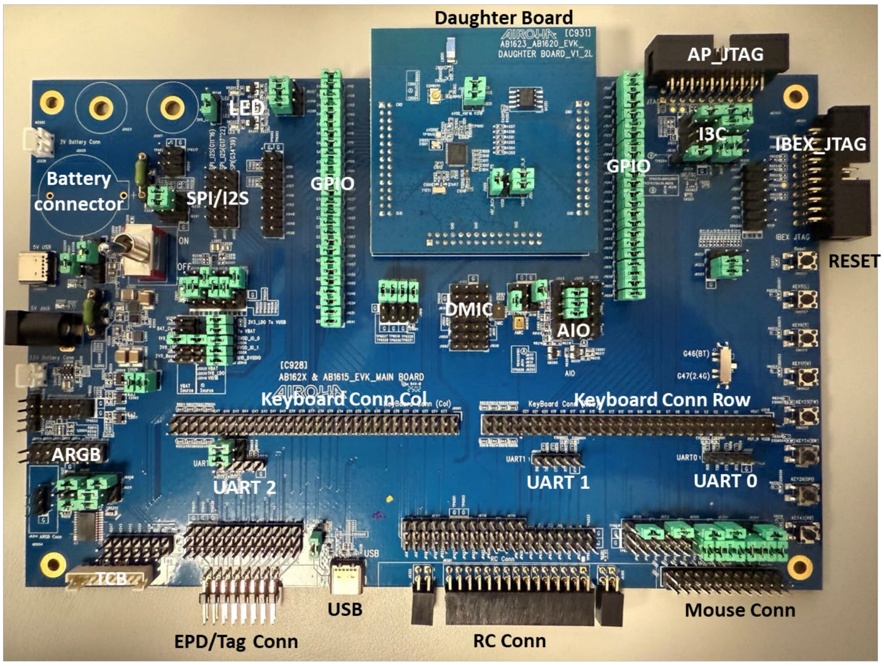

Figure 2-1. Top View of EVK Board

Table 1. Interfaces of EVK

Main Board	Daughter Board
DC Adapter - 5V DC supply	Airoha Bluetooth Chipset
Rock Switch – Switch DC Supply	Antenna – 2.4GHz chip antenna
USB Type-C Connector – VBUS Supply	
Battery Connector – Li-ion / Dry / Coin Battery	
UART Connector – UART 0 / 1 /2	
TCB Connector – TCB (MP Tool) calibration	
Microphone	
SPI Interface – MST0, MST1, MST2 and SLV0	
I2S Interface	
I2C / I3C Interface	
ARGB Interface	
LEDS	
Key Button – Reset and GPIO	
Remote Control Connector	
EPD / Tag Connector	
Mouse Connector	
Keyboard Connector	

3 System Block Diagram

Figure 3-1 shows a block diagram of the EVK. The Airoha chipset is the primary controller. It provides microphone / I2S / UART / I2C / I3C and SPI communication, an I/O control, etc.

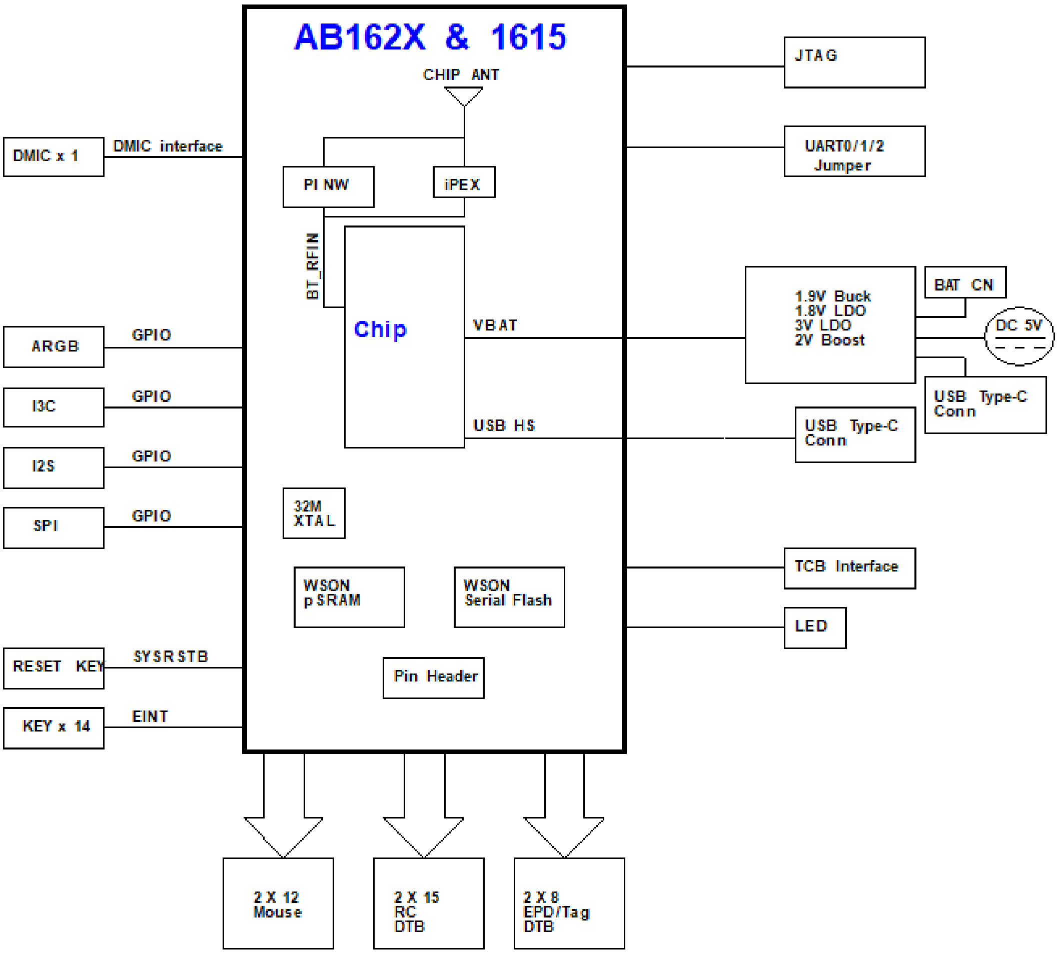

Figure 3-1. EVK Block Diagram

4 Function Blocks

Function blocks of the EVK can be separated into three separate function groups: Peripheral Connections; Control / Test Points; and the Audio Interfaces. These function groups can be separated into more specific function blocks, as shown in Figure 4-1. This document shows how to operate each function block.

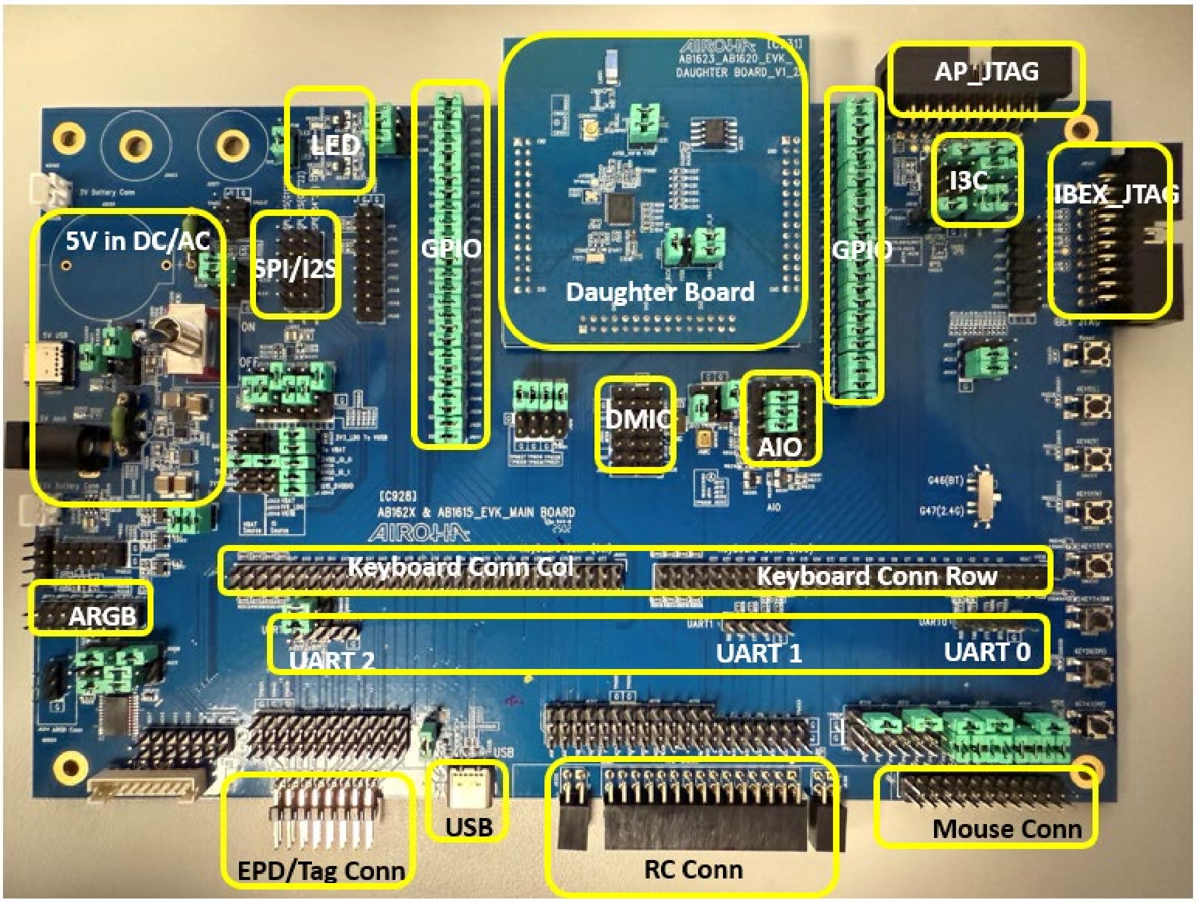

Figure 4-1. Function Blocks of EVK

4.1 Peripheral Connections

4.1.1 Power Supply

There are two types of power supply for the EVK: (1) an external DC power source; the source can be switched from DC Jack or USB Type-C by J2002; (2) a battery, which can be a coin-cell battery, Li-ion battery, or alkaline battery. Alternatively, a battery can also be attached to J2024 / J2025 / J2026 to function as the power supply. Refer to Table 2 to configure the corresponding power supply.

Table 2. Jumper Settings for the Main Board during the Beginning of Application Development

Power Plan	VBAT Volt.	SW2001	J2002	J2019 J2022	J2028	J2031	J2032	J2033	J2034	J2047
Li-ion battery -> external Buck -> AB162x	1.9v	X	X	2-3	X	V	X	X	X	X
Li-ion battery -> external LDO -> AB162x	3.0v	X	X	2-3	X	X	V	X	X	X
Adapter -> external LDO -> AB162x	3.0v	V	2-3 (Adapter) 1-2 (USB CON2001)	1-2	X	X	V	X	X	X
USB(CON2002) -> external LDO -> AB162x	3.3v	X	X	2-3	X	X	X	V	X	V
Coin-cell -> AB162x	1.9~3.3v	X	X	2-3	V	X	X	X	X	X
Alkaline battery x2 -> AB162x	1.9~3.3v	X	X	2-3	V	X	X	X	X	X
Alkaline battery -> Ext. boost -> AB162x	2.0v	X	X	2-3	X	X	X	X	V	X

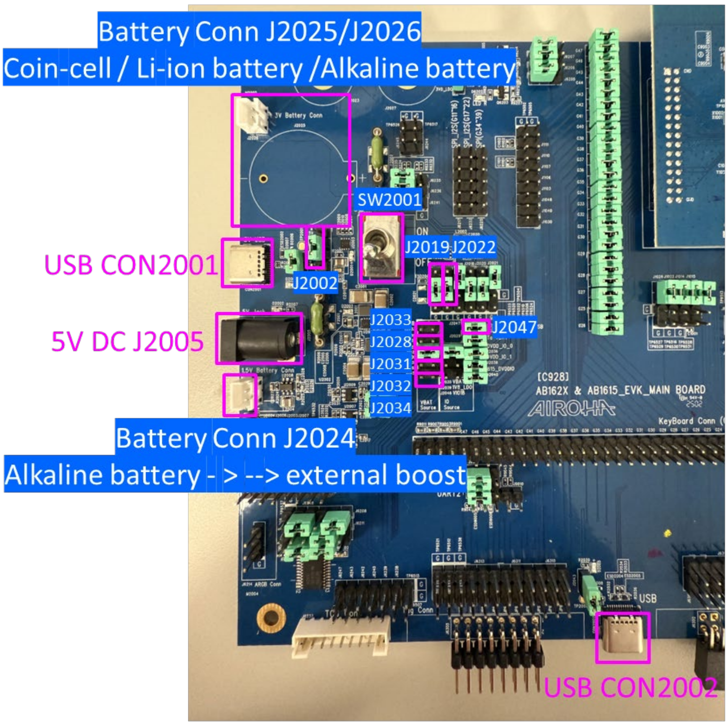

Figure 4-2. Power Supply

4.2 Control and Digital I/O

4.2.1 Module Adaptor Board

Various chipsets and modules can functionally operate with the same EVK motherboard (C928). Each EVK is correctly configured when first distributed.

4.2.2 Key

Figure 4-3 shows the key allocation. They are TACT switches, which pull low each GPIO to the ground when pressed. When the RESET_N key is pressed, it pulls low the RESET_N pin and resets the device.

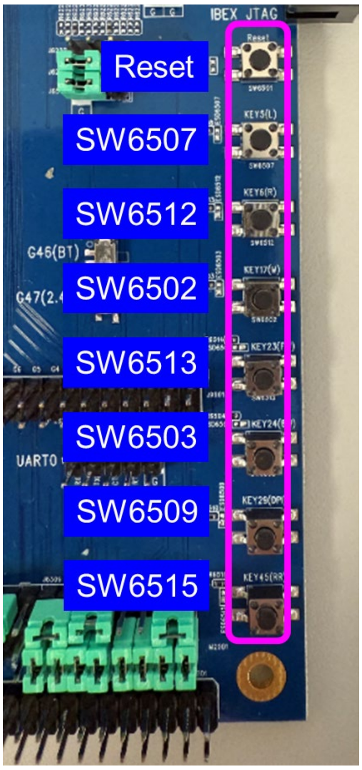

Figure 4-3. Keys

4.2.3 I2C / I3C

There is a set of I2C / I3C interface test points reserved on the EVK. Figure 4-4 shows the location of these test points. The user can use the test points to connect an external I2C slave device and verify the functionality in integration.

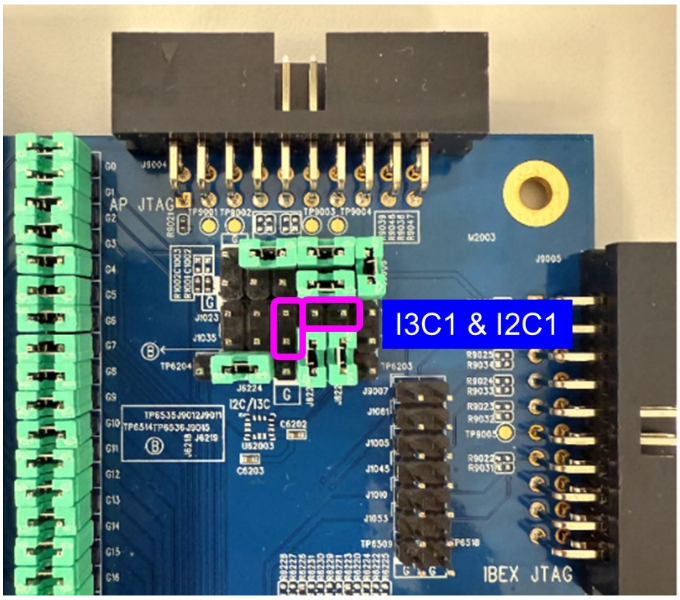

Figure 4-4. I2C / I3C Interface Test Points

4.2.4 UART

There are three sets of reserved UART connectors on the EVK. Figure 4-5 shows the UART connector schematic for the GPIO selection.

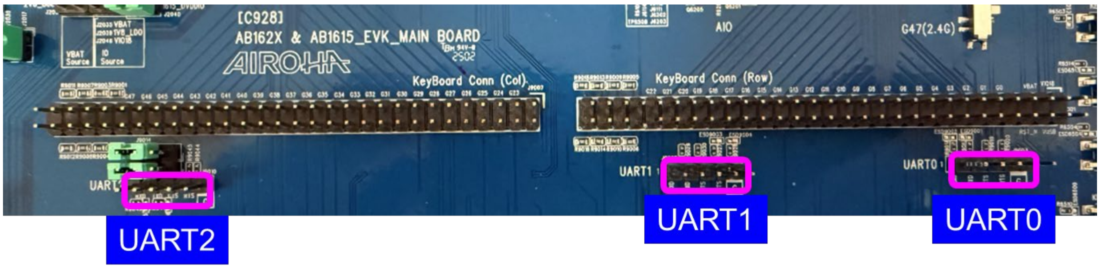

Figure 4-5. UART Interface Test Points

4.2.5 SPI Interface

There are two sets of reserved SPI interface test points on the EVK. Figure 4-6 shows the locations of these test points. Users can use these test points to verify the functionality during integration.

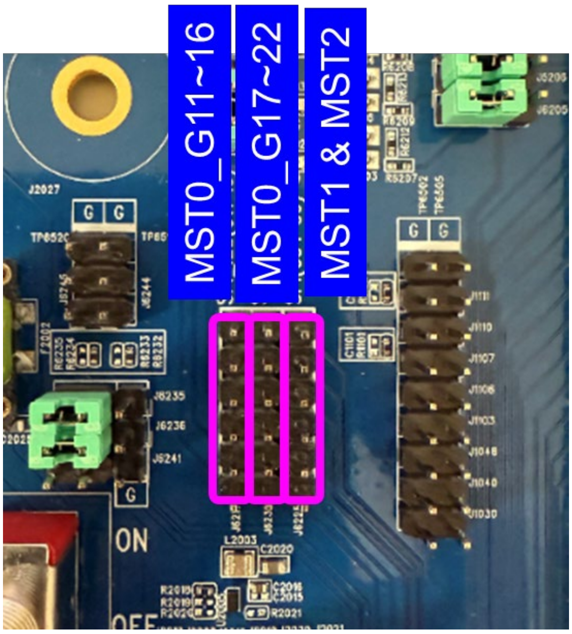

Figure 4-6. SPI Interface Test Points

4.2.6 GPIO

Figure 4-7 shows all the GPIO test pins. These pins can be used to evaluate functions.
There might be some additional electrical components or circuits connected to the GPIO that can be used. Refer to the EVK schematic to disconnect these jumpers.

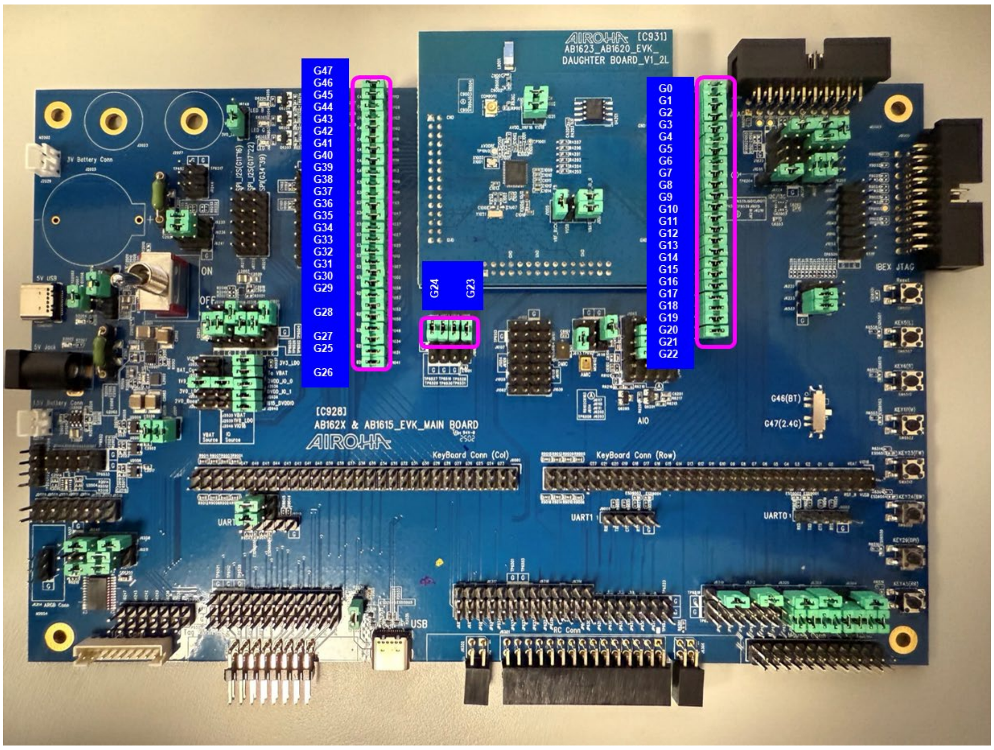

Figure 4-7. All GPIO Test Points

4.2.7 TCB

TCB is a calibration kit for adjusting the crystal of an Airoha chipset. There is a built-in TCB connector J917 on the EVK. Figure 4-8 shows the location of the TCB connector. J6237 is the default connector for connecting TCB. J6247, J6243, J6242, J6240, J6239, and J6238 (described in the 4.1.1 Power Supply and 4.2.4 UART sections) must connect during TCB calibration.

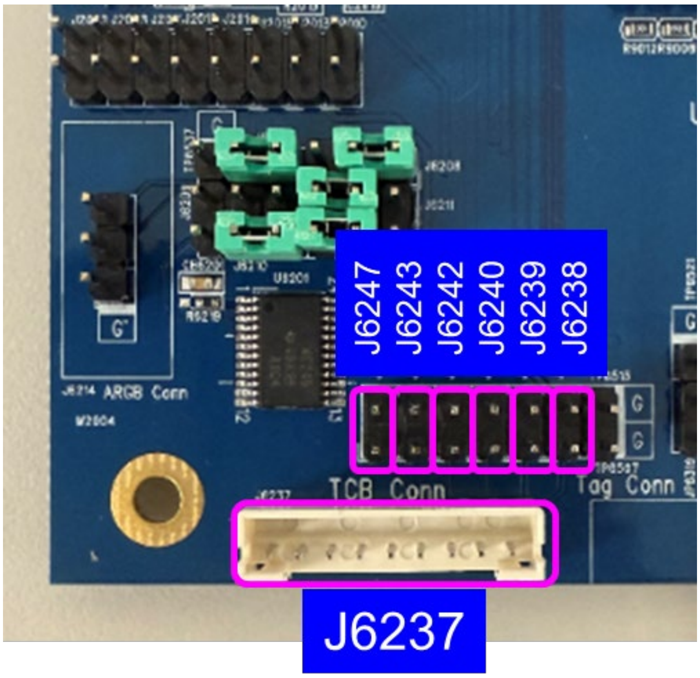

Figure 4-8. TCB Connector

4.2.8 FSOURCE_D

FSOURCE_D serves as an eFUSE write power source. To write eFUSE, provide VIO18 voltage to the FSOURCE_D test point on the daughter board.

4.2.9 LEDs

The LEDs are connected to the GPIO pins on AB162x. The user can program and evaluate LED behaviors through these three LEDs. Figure 4-9 and Figure 4-10 show the connection of the debugging LEDs.

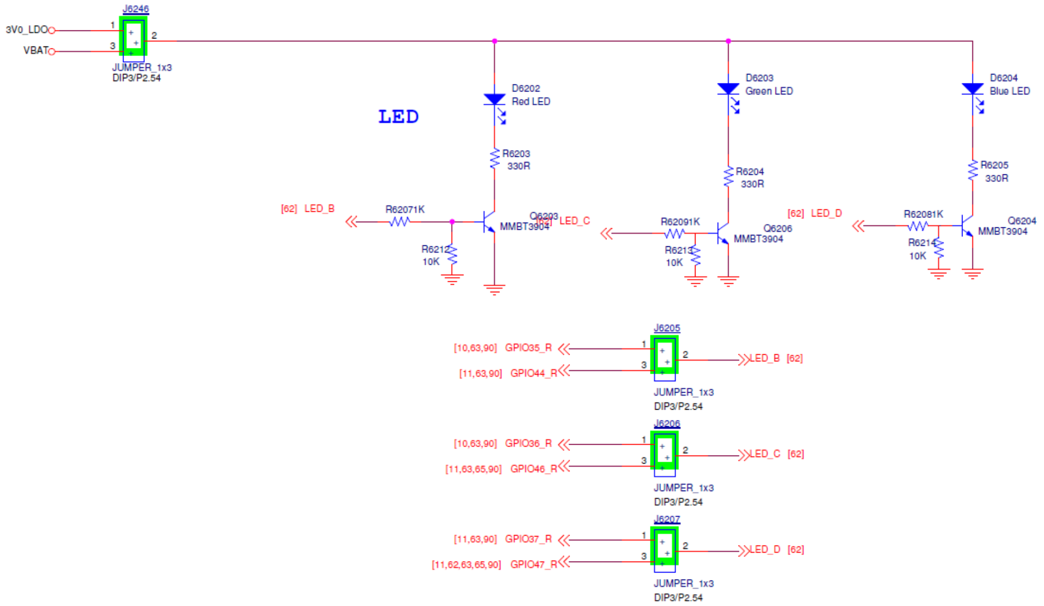

Figure 4-9. Schematics of LED Circuit

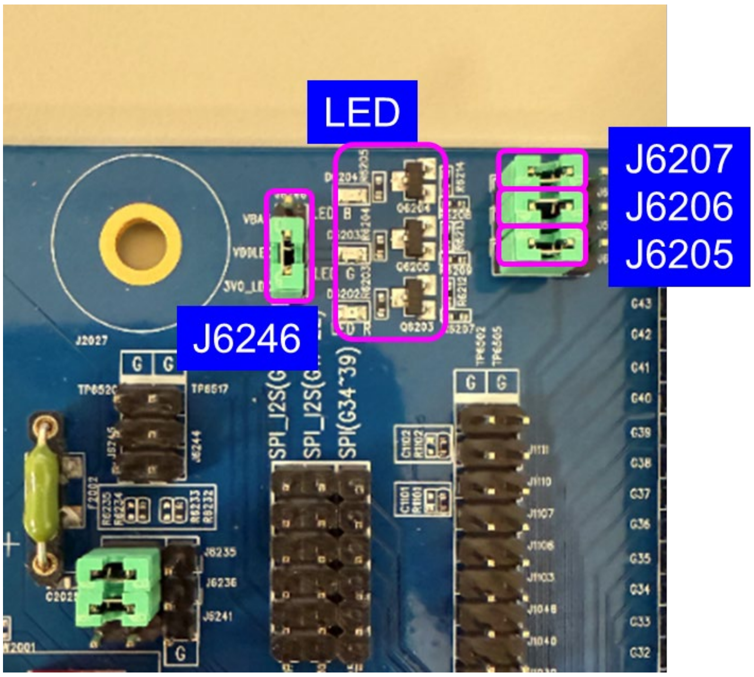

Figure 4-10. LED Locations

4.2.10 ARGB

The ARGB connects to the GPIO pins on AB162x. GPIO34 or GPIO43 can be selected using J6209.

Figure 4-12 shows the locations of these test points.

Adjust the following jumpers to change the operating voltage of ARGB: J6209, J6210, J6211, J6213, J6208, and J6248.

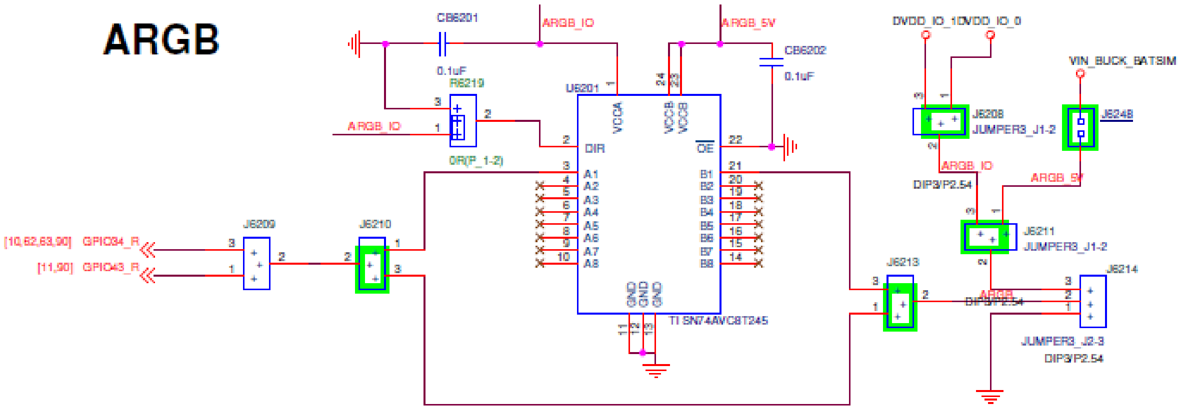

Figure 4-11. Schematics of ARGB Circuit

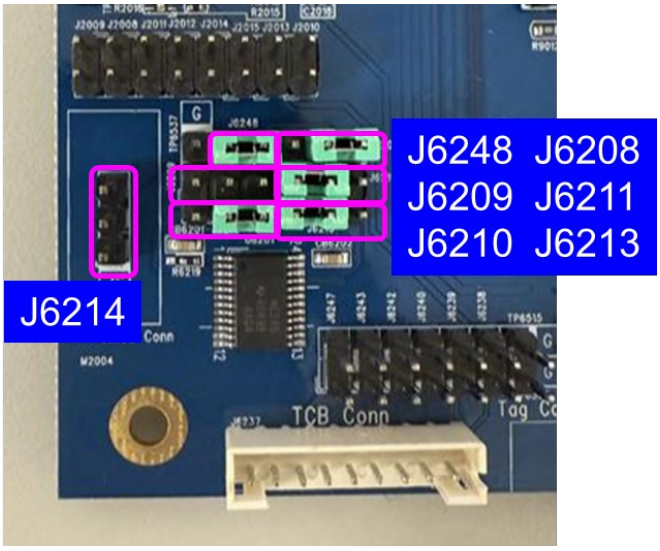

Figure 4-12. ARGB Locations

4.2.11 AuxADC / AIO

AB162x utilizes several GPIO pins to support the voltage detection function. Figure 4-14 shows J6202 and J6203.

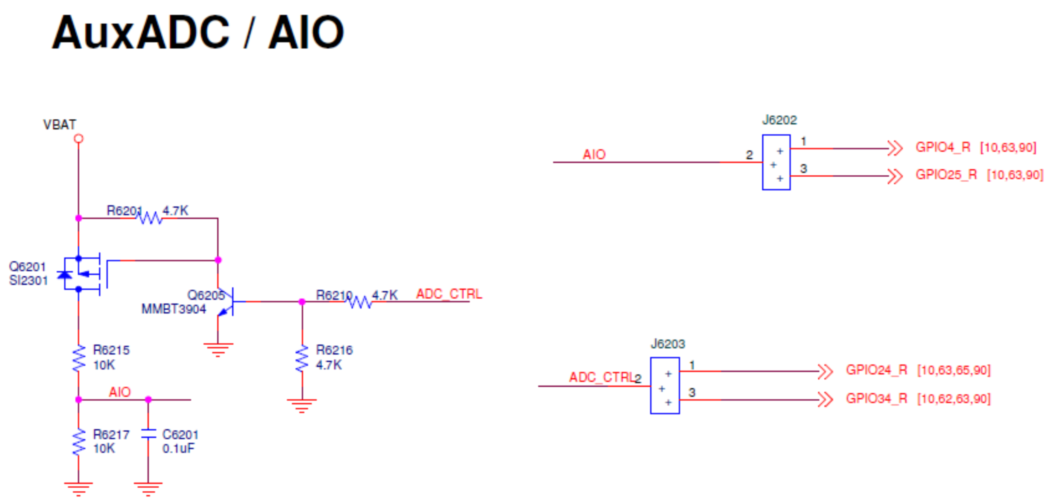

Figure 4-13. Schematics of AuxADC Circuit

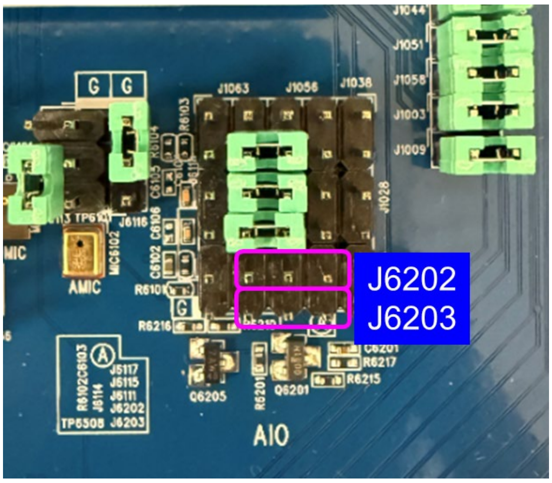

Figure 4-14. AuxADC Location

4.2.12 Mouse Connector

This pin header (CON6301) integrates applications commonly found in the standard mouse. The user can use the actual mouse mechanism connected to the AB162x EVK and test the mouse feature.

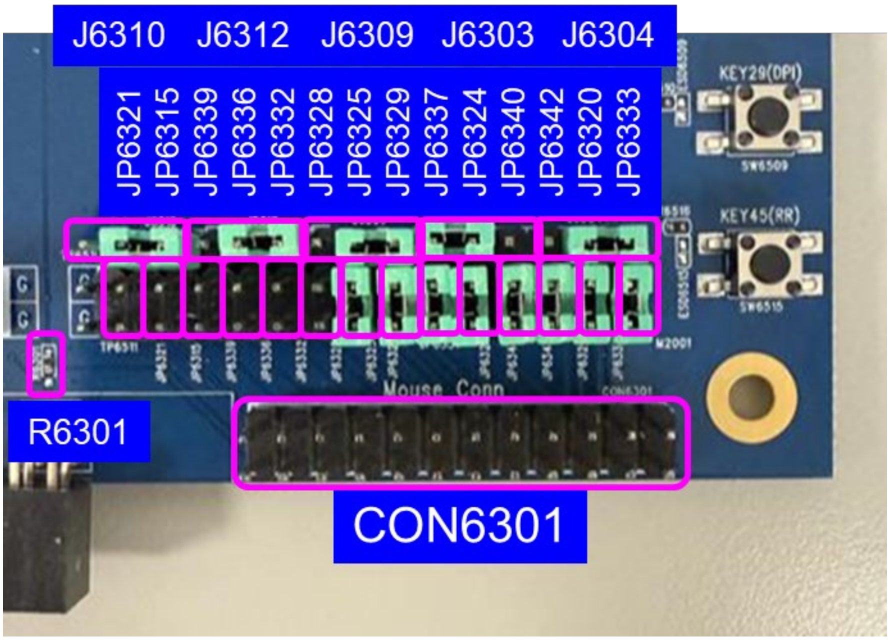

Figure 4-15. Mouse Connector and Jumper Locations

4.2.13 Remote Control Daughter Board Connector

The remote-control daughter board pin headers can be configured in user-defined applications. The user can use the remote-control board to connect to the AB162x EVK and test use the matrix keypad, IR LED, Li-ion battery, and gyro / accelerometer sensor.

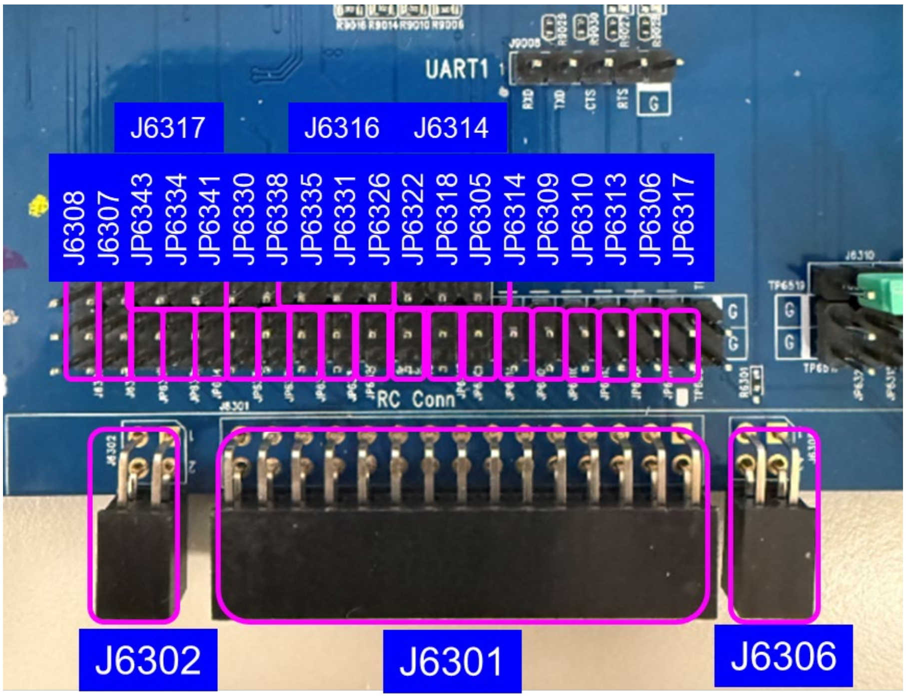

Figure 4-16. Remote Control Daughter Board Connector and Jumper Locations

4.2.14 EPD / Findmy Tag Connector

This pin header (J6305) integrates applications commonly found in the EPD & Findmy Tag. The user can use the actual EPD & Findmy Tag mechanism connected to the AB162x EVK and test related features.

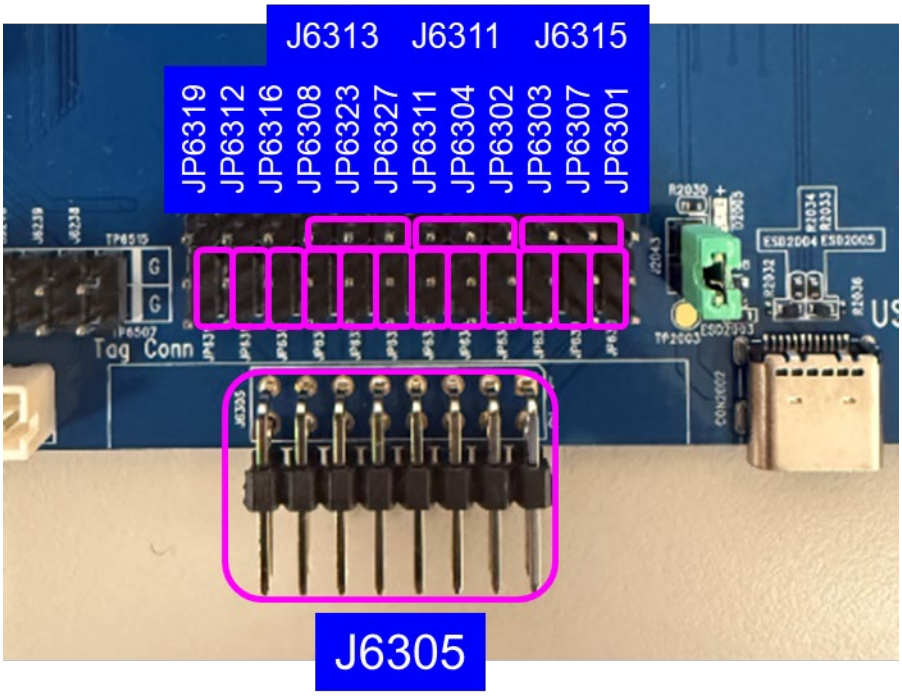

Figure 4-17. EPD & Findmy Tag Connector and Jumper Locations

4.3 Audio Interface

4.3.1 Microphone

There is one digital microphone on the AB162x EVK. Figure 4-18 and Figure 4-19 show the schematics of the digital microphone circuit and the jumper locations.

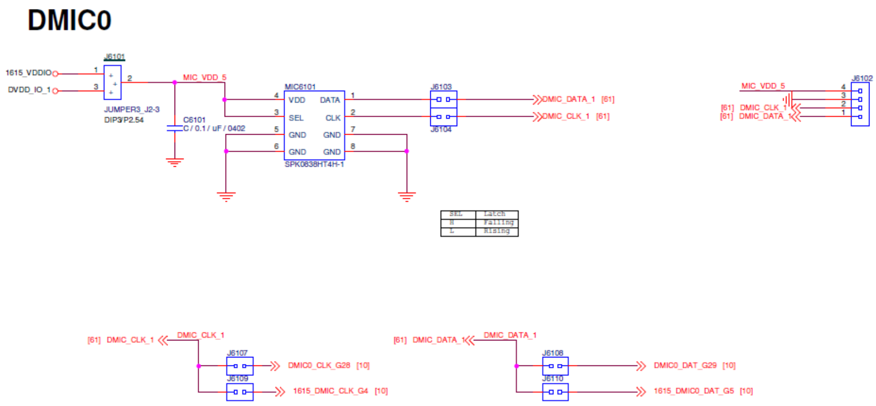

Figure 4-18. Schematics of Digital Microphone Circuit

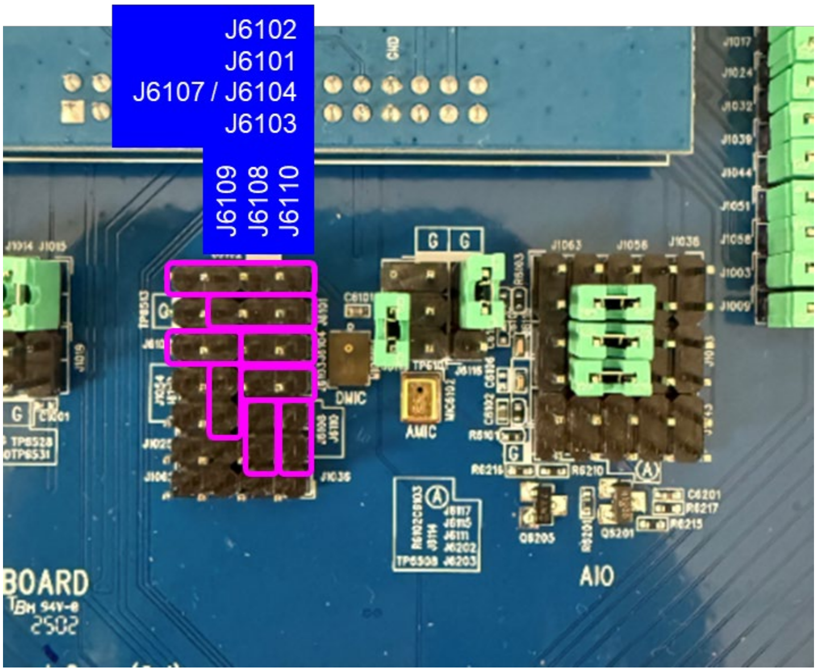

Figure 4-19. Location of Jumper

4.3.2 I2S

AB162x chipsets support a set of I2S interfaces. J6217 / J6232 are located at the left-side of the EVK motherboard, as shown in Figure 4-20. Figure 4-20 also shows the functions of each pin.

Table 3. I2S GPIO Option

	Pin Name	I2S_MSTx_WS	I2S_MSTx_RX	I2S_MSTx_TX	I2S_MSTx_CK
I2S MST0_1	GPIO	GPIO14	GPIO12	GPIO13	GPIO11
I2S MST0_2	GPIO	GPIO22	GPIO20	GPIO21	GPIO19

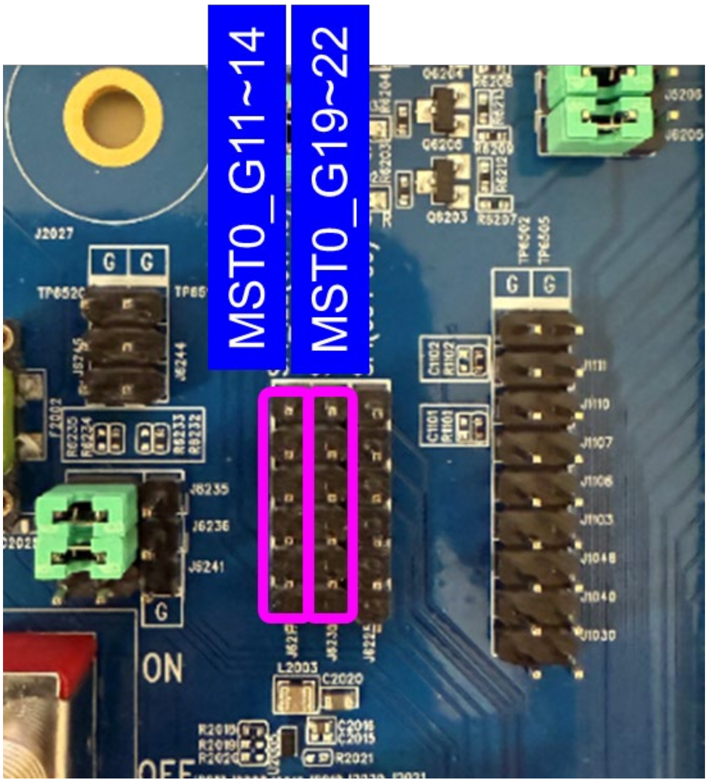

Figure 4-20. Two sets of I2S Interface

5 Downloading the Project Image Using the EVK

The Airoha IoT Flash Tool downloads the project image. The link for downloading the Airoha IoT Flash Tool is available via the MOL portal.
The following section describes the procedure to download the project file using the Airoha IoT Flash Tool.

5.1 Downloading the Firmware with UART / USB

To download the firmware with UART / USB, complete the following steps:

1. Start Airoha IoT Flash Tool and open the image `*.cfg` file.

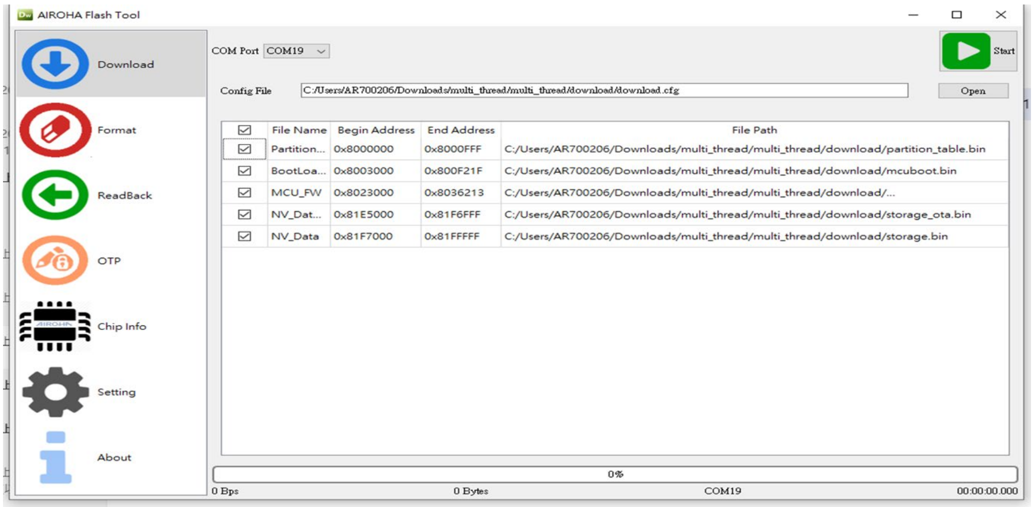

Figure 5-1. Opening the *.cfg File in Airoha IoT Flash Tool

2. In the device manager (Figure 5-2), confirm the correct UART / USB COM.

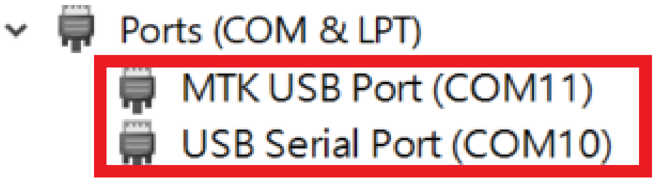

Figure 5-2. Device Manager

3. Make sure the **COM Port** is set to “UART” or “USB” and then click `Start`. See Figure 5-3. This stops the Airoha IoT Flash Tool.

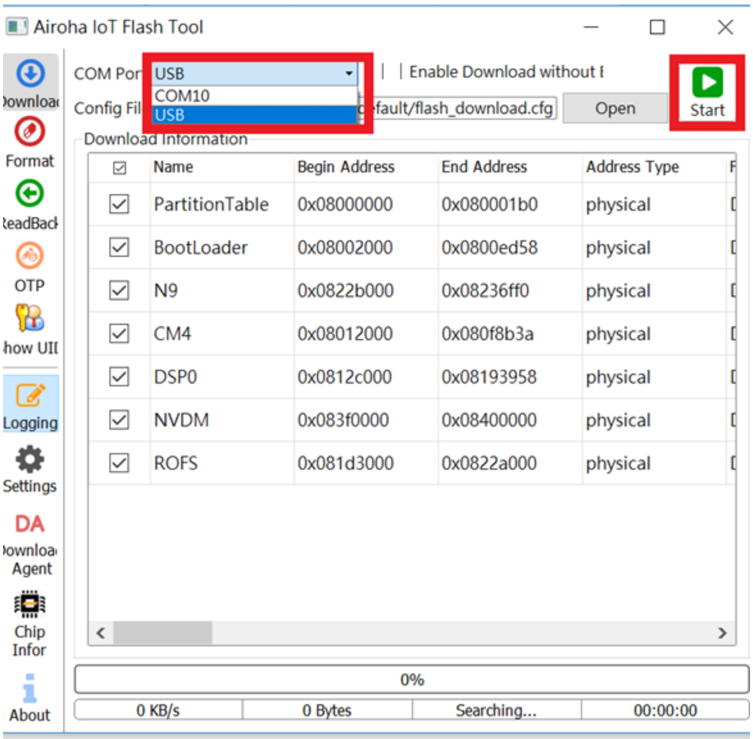

Figure 5-3. Airoha IoT Flash Tool

4. Set the UART / USB on the board, as shown in Figure 5-4.
5. Use the UART / USB cable to connect the EVK UART0 / USB and then click the EVK `Reset` button to start the download process. The download process is displayed by the Airoha IoT Flash Tool.

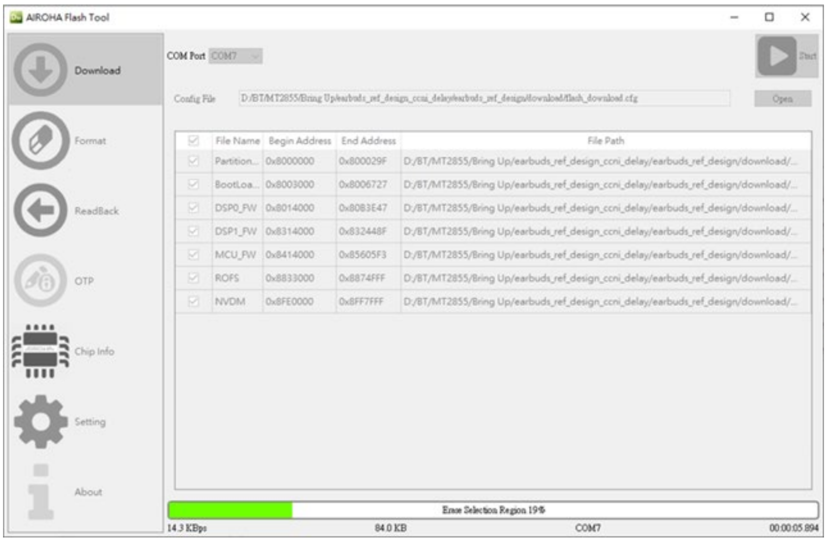

Figure 5-4. Downloading the Flash File

6. The Airoha IoT Flash Tool shows "Success" when the download is complete. Click any button to close the window.

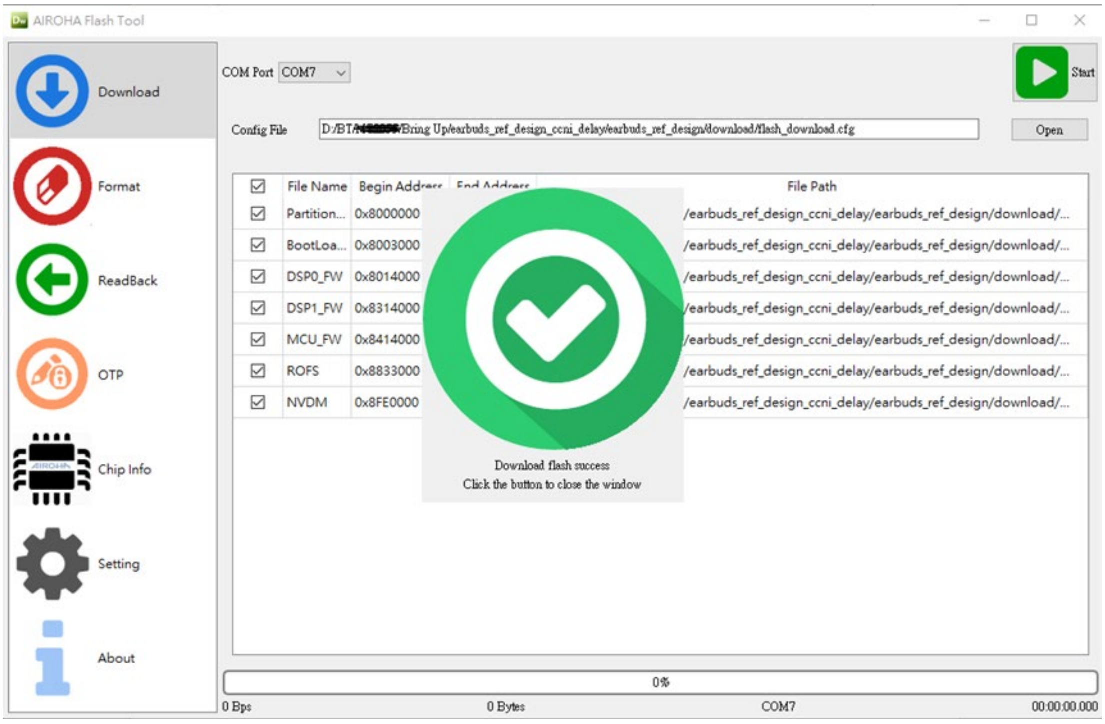

Figure 5-5. Flash Download Success Notification

Exhibit 1 Terms and Conditions

Your access to and use of this document and the information contained herein (collectively this "Document") is subject to your (including the corporation or other legal entity you represent, collectively "You") acceptance of the terms and conditions set forth below ("T&C"). By using, accessing or downloading this Document, You are accepting the T&C and agree to be bound by the T&C. If You don't agree to the T&C, You may not use this Document and shall immediately destroy any copy thereof.

This Document contains information that is confidential and proprietary to Airoha Technology Corp. and/or its affiliates (collectively "Airoha") or its licensors and is provided solely for Your internal use with Airoha's chipset(s) described in this Document and shall not be used for any other purposes (including but not limited to identifying or providing evidence to support any potential patent infringement claim against Airoha or any of Airoha's suppliers and/or direct or indirect customers). Unauthorized use or disclosure of the information contained herein is prohibited. You agree to indemnify Airoha for any loss or damages suffered by Airoha for Your unauthorized use or disclosure of this Document, in whole or in part.

Airoha and its licensors retain titles and all ownership rights in and to this Document and no license (express or implied, by estoppels or otherwise) to any intellectual propriety rights is granted hereunder. This Document is subject to change without further notification. Airoha does not assume any responsibility arising out of or in connection with any use of, or reliance on, this Document, and specifically disclaims any and all liability, including, without limitation, consequential or incidental damages.

THIS DOCUMENT AND ANY OTHER MATERIALS OR TECHNICAL SUPPORT PROVIDED BY AIROHA IN CONNECTION WITH THIS DOCUMENT, IF ANY, ARE PROVIDED "AS IS" WITHOUT WARRANTY OF ANY KIND, WHETHER EXPRESS, IMPLIED, STATUTORY, OR OTHERWISE. AIROHA SPECIFICALLY DISCLAIMS ALL WARRANTIES OF MERCHANTABILITY, NON-INFRINGEMENT, FITNESS FOR A PARTICULAR PURPOSE, COMPLETENESS OR ACCURACY AND ALL WARRANTIES ARISING OUT OF TRADE USAGE OR OUT OF A COURSE OF DEALING OR COURSE OF PERFORMANCE. AIROHA SHALL NOT BE RESPONSIBLE FOR ANY AIROHA DELIVERABLES MADE TO MEET YOUR SPECIFICATIONS OR TO CONFORM TO A PARTICULAR STANDARD OR OPEN FORUM.

Without limiting the generality of the foregoing, Airoha makes no warranty, representation or guarantee regarding the suitability of its products for any particular purpose, nor does Airoha assume any liability arising out of the application or use of any product, circuit or software. You agree that You are solely responsible for the designing, validating and testing Your product incorporating Airoha's product and ensure such product meets applicable standards and any safety, security or other requirements.

The above T&C and all acts in connection with the T&C or this Document shall be governed, construed and interpreted in accordance with the laws of Taiwan, without giving effect to the principles of conflicts of law.